

# Design Traffic

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## Role of Traffic



### ▪ Geometric Design

- An account of all traffic volume should be considered to come up with the lane width, horizontal curves, and other design elements.

### ▪ Pavement Design

- Only heavy vehicles "Heavy Commercial Vehicles) is considered in the pavement structural design and the damage analysis.

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# Pavement Damage

- The damage caused to a pavement by the passage of a heavy vehicle depends not only on its gross weight but also on how this weight is distributed to the pavement. This damage depends on:
  - ❖ The number of axles in the vehicle,
  - ❖ The manner in which these axles are grouped together into axle group; and
  - ❖ The load applied to the pavement by each of these axle groups.

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## Different Axle Groups (Configurations)

- ❖ Single Axle with Single Tyres (SAST);
- ❖ Single Axle with dual Tyres (SADT);
- ❖ Tandem Axle with Single Tyres (TAST);
- ❖ Tandem Axle with Dual Tyres (TADT);
- ❖ Triaxle (Tridem) with Dual Tyres (TRDT);
- ❖ Quad-Axle with Dual Tyres (QADT).

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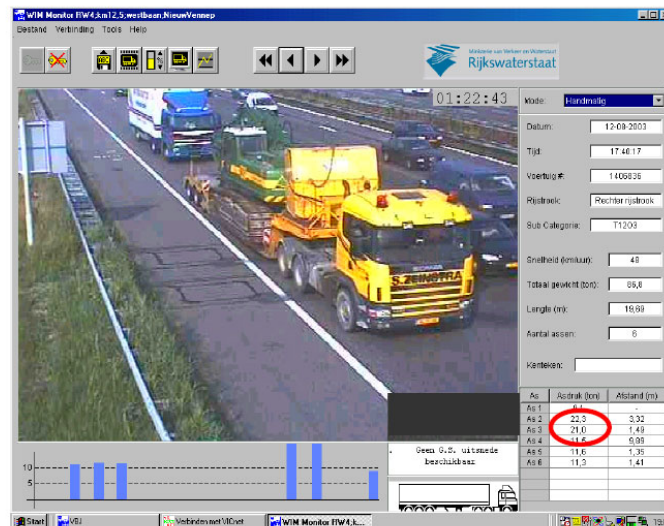


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# Weigh in Motion



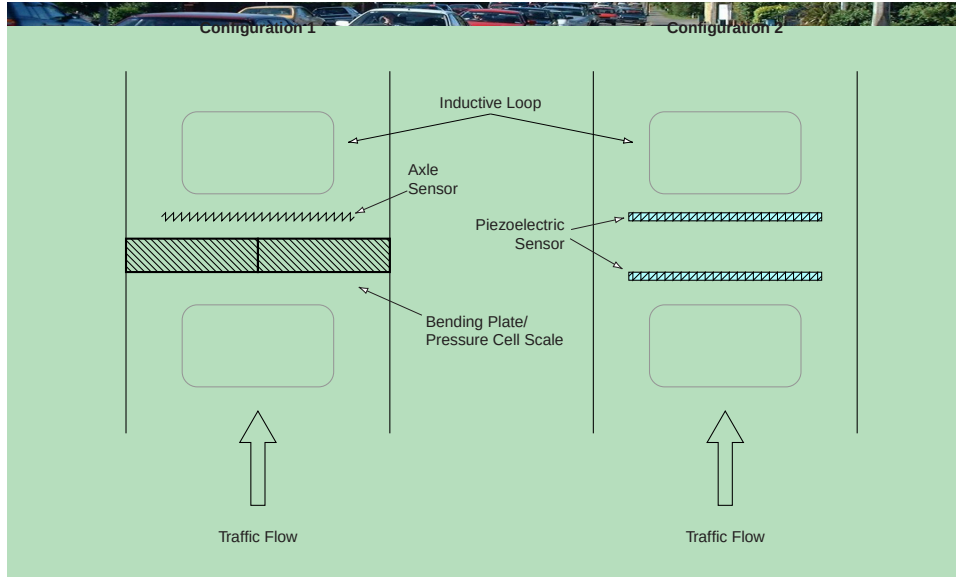
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# Cumulative Traffic

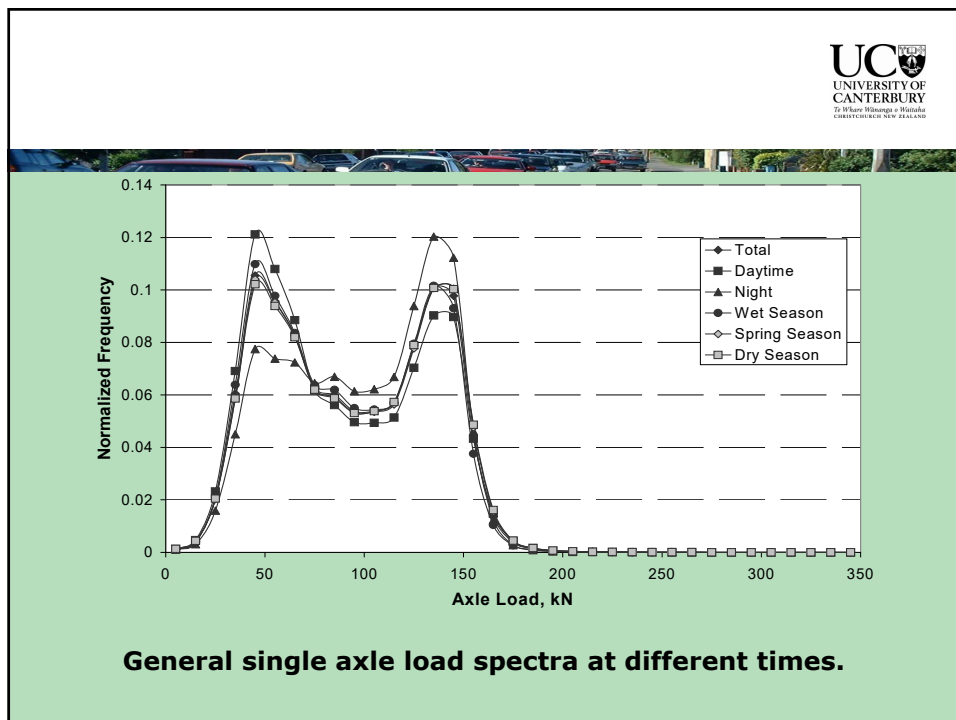
- Traffic loading is computed as the cumulative number of each axle loading (by type and weight) traversing the pavement within the design period.
- This cumulative loading is described by:
  1. Cumulative number of axle groups traversing the pavement during the period;
  2. The proportion of each axle group in this total; and
  3. For each axle group type, the frequency distribution of the axle group loads.

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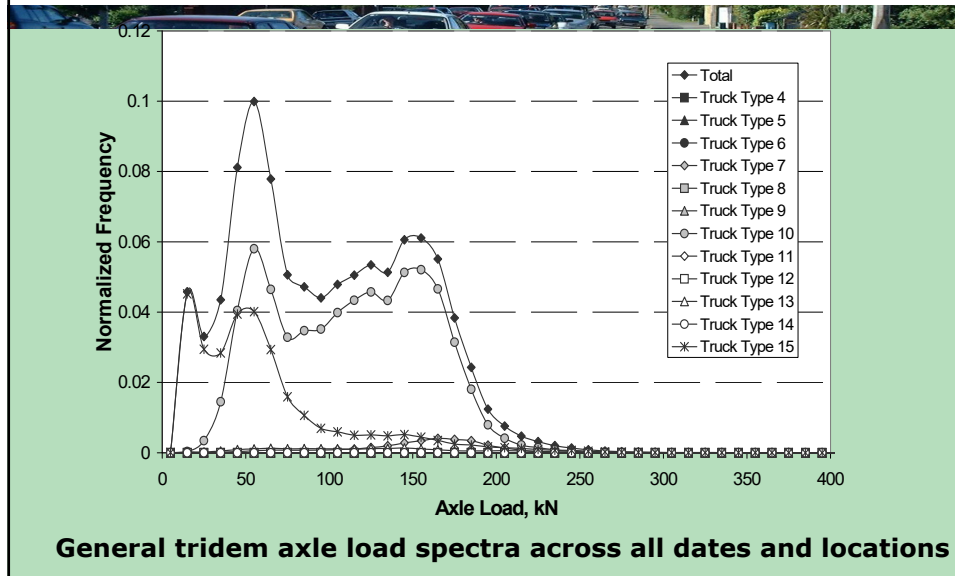
# Weigh in Motion (WIM)



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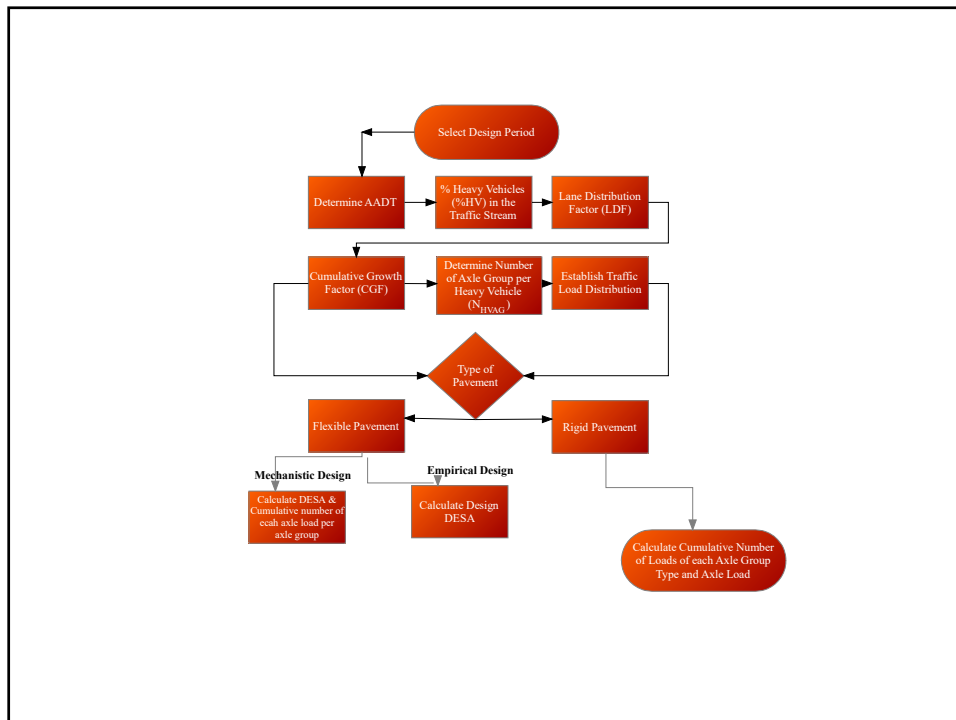


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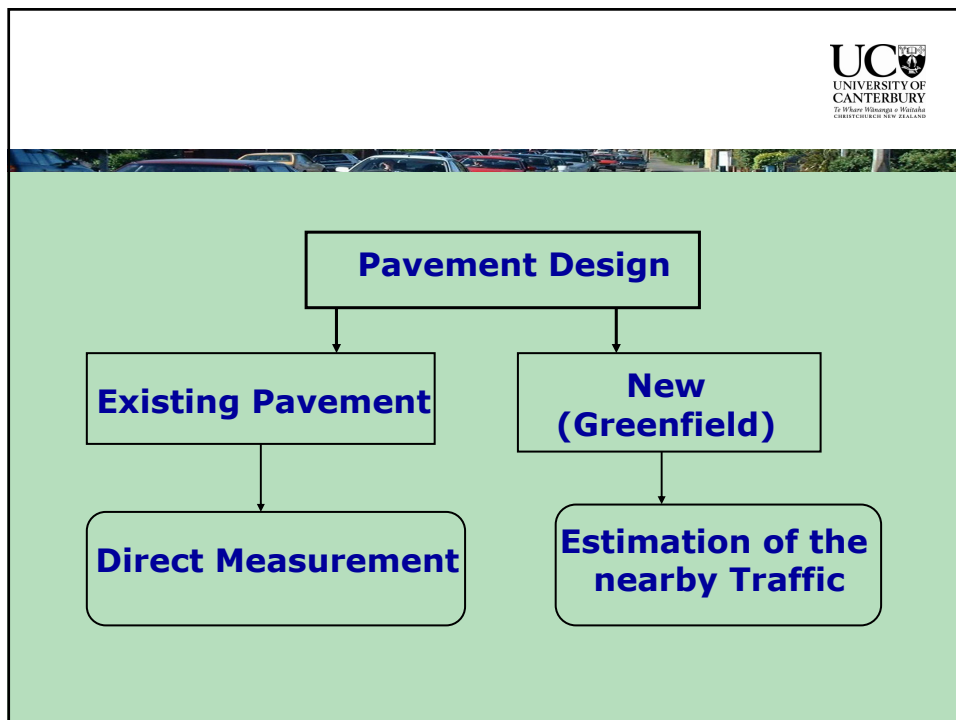
## Traffic Load Distribution (TLD)

| Axle Group |      |       |       |       |      |       |      |
|------------|------|-------|-------|-------|------|-------|------|
| Load(kN)   |      | SAST  | SADT  | TAST  | TADT | TRDT  | QADT |
| 0          | 10   | 2.55  | 3.28  |       |      | 2.5   |      |
| 10         | 20   | 16.17 | 13.11 |       |      | 3.98  | 0.64 |
| 20         | 30   | 22.13 | 26.23 |       |      | 6.97  | 1.64 |
| 30         | 40   | 14.89 | 24.59 |       |      | 7.46  | 5.1  |
| 40         | 50   | 28.94 | 12.3  | 33.33 |      | 9.45  | 12.1 |
| 50         | 60   | 11.06 | 9.84  | 33.33 |      | 5.97  | 12.1 |
| 60         | 70   | 2.98  | 5.74  |       |      | 8.46  | 8.92 |
| 70         | 80   | 1.28  | 2.46  |       |      | 12.44 | 9.55 |
| 80         | 90   |       | 1.64  | 16.67 |      | 10.45 | 8.92 |
| 90         | 100  |       | 0     |       |      | 7.46  | 9.55 |
| 100        | 110  |       | 0.81  |       |      | 6.47  | 7.64 |
| 110        | 120  |       |       | 16.67 |      | 4.98  | 4.46 |
| 120        | 130  |       |       |       |      | 5.47  | 3.82 |
| 130        | 140  |       |       |       |      | 3.48  | 2.8  |
| 140        | 150  |       |       |       |      | 2.47  | 6.36 |
| 150        | 160  |       |       |       |      | 1.99  | 6.4  |
|            |      | 100   | 100   | 100   | 100  | 100   | 100  |
| SAST       | SADT | TAST  | TADT  | TRDT  | QADT |       |      |
| 32.55      | 16.9 | 0.83  | 27.84 | 21.75 | 0.13 | 100   |      |

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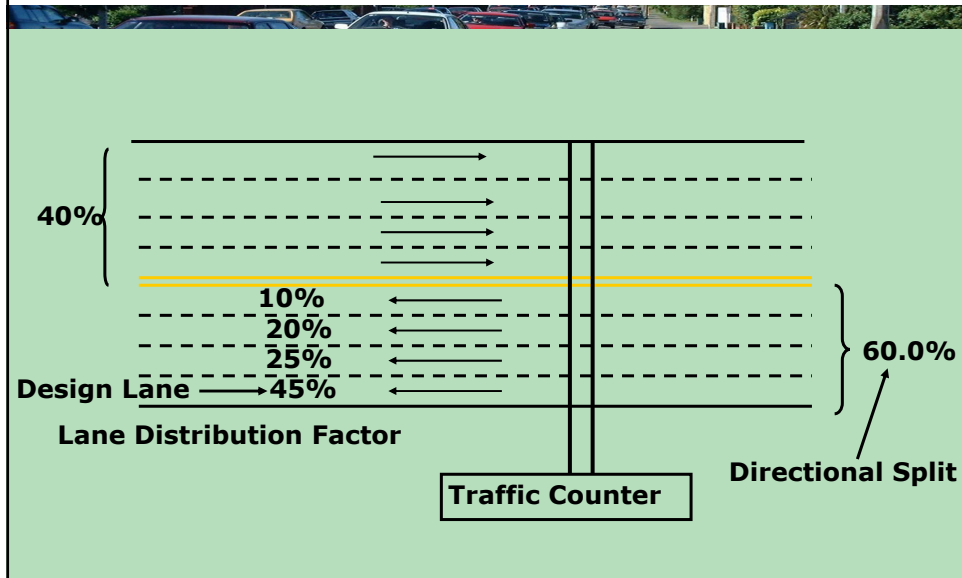
## Design Traffic calculations based on AADT

- Select a design period based on pave type
- Identify the most heavily-trafficked lane in the heaviest trafficked direction “design lane”. Estimate the average daily number of heavy vehicles in the design lane during the first year of the project’s life
- Estimate heavy traffic growth throughout the design period ( $r$ ).
- Estimate the average number of axle groups per heavy vehicle  $N_{HVAG}$ .

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- Combine the above three estimates to calculate the cumulative heavy vehicle axle groups over the design period  $N_{DT}$ .
- Estimate the proportion of axle group types and the distribution of axle group loads.
- Express the cumulative traffic loading in a form suitable for the pavement design procedure to be used

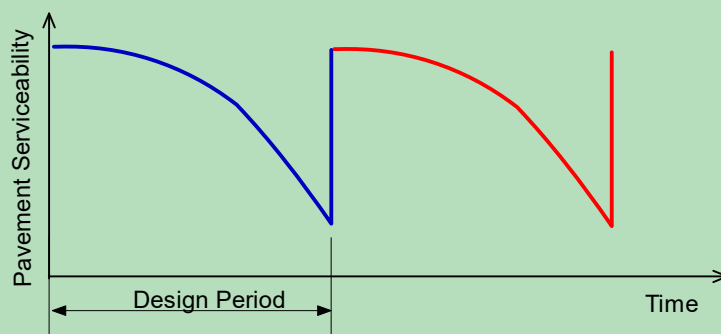
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## Design Period

- Design Period is the length of time considered appropriate for the road pavement to function without major rehabilitation or reconstruction.



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## Factors to be Considered

- Available funds for the project
- Hierarchical rank of the road (Importance)
- Likelihood of future realignment (e.g to improve Geometric standard)
- Likelihood of future upgrade (e.g to improve the structural capacity of the road).
- Other factors (e.g. reactive subgrades, consolidation of imported fills or compressible soil strata) that will cause distresses necessitating major rehabilitation.
- The likelihood of existing fixed levels (kerb and gutter, clearance under overhead structures, etc.) constraining the selection of rehabilitation treatments to more costly options.

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## Design Period (Cont.)

| Type of Pavement   | Design Periods (Years) |
|--------------------|------------------------|
| Flexible Pavements | 20-30                  |
| Rigid Pavements    | 30-40                  |

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## Design Lane

- It is the most heavily trafficked lane.
- The distribution of traffic across the lanes is dependent on:
  - The number of traffic lanes
  - The presence of parked vehicles in the left lane
  - The proximity of intersections; and
  - Primary and secondary functions of the road.

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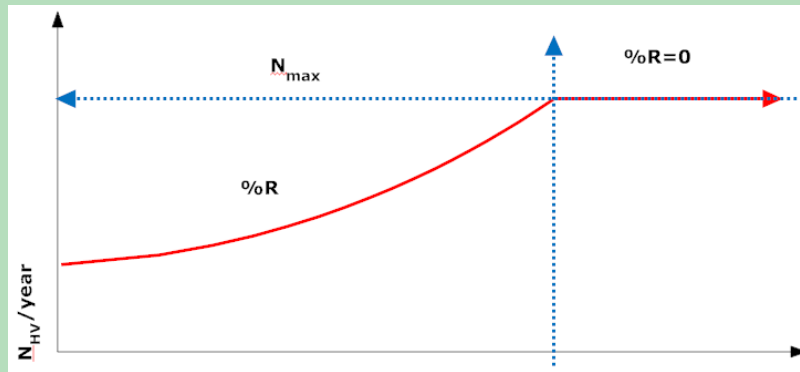
## Design Lane (Cont.)

- Weigh in motion (WIM) systems can provide lane specific traffic loading data across a carriageway. In the absence of project-specific data, Table 1 can provide guidance.

| Location Lanes/direction |         | L Lane | C Lane | R Lane |
|--------------------------|---------|--------|--------|--------|
| Rural                    | 2 lanes | 1.00   | N/A    | 0.50   |
|                          | 3 lanes | 0.95   | 0.65   | 0.30   |
| Urban                    | 2 lanes | 1.00   | N/A    | 0.50   |
|                          | 3 lanes | 0.65   | 0.65   | 0.50   |

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## Heavy Vehicles < Lane capacity



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## Lane capacity in terms of passenger cars (Aust 2017)

Table 7.5: Capacity flow rates

| Road conditions                                                                  | Capacity flow rate<br>(passenger cars per hour per lane) |
|----------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Motorway</b>                                                                  |                                                          |
| Through lanes                                                                    | 2300                                                     |
| Ramps                                                                            | 2000                                                     |
| <b>Multilane road (two or more lanes in each direction, other than motorway)</b> |                                                          |
| posted speed 100 km/h                                                            | 2200                                                     |
| posted speed 90 km/h                                                             | 2100                                                     |
| posted speed 80 km/h                                                             | 2000                                                     |
| posted speed 70 km/h                                                             | 1900                                                     |
| posted speed <70 km/h                                                            | 1850                                                     |
| Two-lane, two-way (one lane in each direction)                                   | 1700                                                     |
| Signalised intersections                                                         | 1850 <sup>(1)</sup>                                      |

<sup>1</sup> For signalised intersections, this flow rate is multiplied by green time proportion (see Austroads 2013b).

Source: Austroads (2013b).

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## Pc equivalents (Aust 2017)

- Austroads vehicle classes 2 to 5:  $E_{HV} = 2$
- Austroads vehicle classes 6 to 9:  $E_{HV} = 3$
- Austroads vehicle classes 10 to 11:  $E_{HV} = 4$
- Austroads vehicle class 12:  $E_{HV} = 5$

$$C_{pce} = \frac{C_{pc}}{\left(1 + \frac{P_{HV}}{100} * (E_{HV} - 1)\right)}$$

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## Adjusted capacity

$$C_{pce} = \frac{C_{pc}}{\left(1 + \frac{P_{HV}}{100} * (E_{HV} - 1)\right)}$$

- $C_{pce}$  = Max hourly volume of passenger car equivalents per lane
- $C_{pc}$  = Max hourly volume of passenger cars per lane
- $P_{HV}$  = % of heavy vehicles
- $E_{HV}$  = average number of passenger cars equivalent to HV

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# Capacity in Terms of HV

$$C_{HV} = \frac{C_{pce}}{\left( E_{HV} + \left( \frac{100}{P_{HV}} - 1 \right) \right)}$$

**$C_{HV}$  = Max hourly number of HVs per lane (HV/h)**

**$C_{pce}$  = Max hourly volume of pc equivalents per lane**

**$E_{HV}$  = average number of passenger car equivalents to a HV**

**$P_{HV}$  = % of HV in the traffic stream**

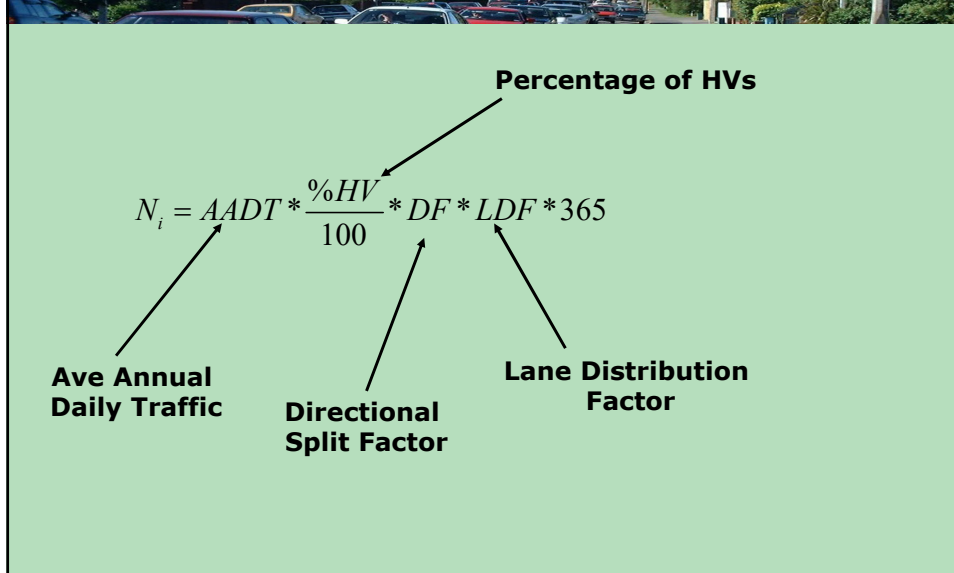
**Lane capacity of heavy vehicles per year can be calculated based on the above equation**

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$$N_{\max} = C_{HV} * 24 * 365$$

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# Determination of the total Heavy Vehicles Per Year



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$$N_{HVn} = N_i * \left(1 + \frac{R}{100}\right)^{n-1}$$

**Solve of n**

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## Lane under capacity for the entire design period

- Traditionally the traffic growth trend is assumed to be geometric in nature, i.e. it can be modelled by the conventional compound growth formulae. The traffic growth is usually expressed as a percentage increase in the annual traffic volume (Annual traffic growth rate is R%).

$$CGF = \frac{\left(1 + \frac{R\%}{100}\right)^n - 1}{\frac{R\%}{100}}$$

**R=0 CGF=n**

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## Design Traffic for Lane Under Capacity

$$N_{DT} = N_i * N_{HVAG} * CGF$$

**N<sub>i</sub>**= Number of Heavy Vehicles per year 1

**N<sub>HVAG</sub>**= Number of Heavy Axle Groups per Heavy Vehicle

**CGF**= Cumulative growth factor

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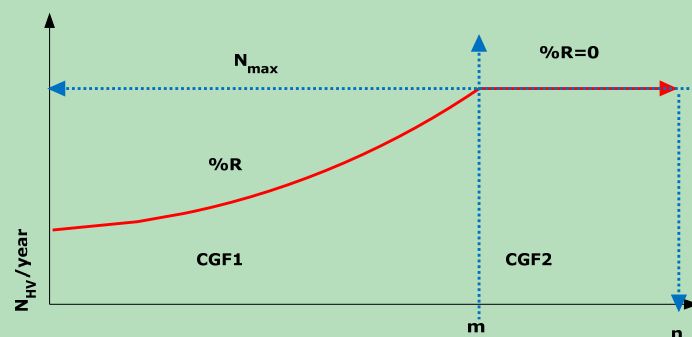
# Operate over Capacity

- In this case adjustment will be needed as follows

Use  $N_{\max}$  to determine the number of years in which the lane is just at capacity; solve for  $m$

$$N_{\max} = N_i * \left(1 + \frac{R}{100}\right)^{m-1}$$

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$$CGF = \underbrace{\frac{\left(1 + \frac{\%R}{100}\right)^m - 1}{\frac{R}{100}}}_{CGF1} + \underbrace{(n - m) * \left(1 + \frac{\%R}{100}\right)^{m-1}}_{CGF2}$$

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## Design Traffic for adjusted HV growth

$$N_{DT} = N_i * N_{HVAG} * CGF$$

**N<sub>i</sub>**= Number of Heavy Vehicles per year 1

**N<sub>HVAG</sub>**= Number of Heavy Axle Groups per Heavy Vehicle

**CGF**= Cumulative growth factor (CGF1+CGF2)

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## Estimating Axle Groups per Heavy Vehicle

- $N_{HVAG}$  should be estimated based on the WIM or the Classification Counter Data. In the absence of such information a presumptive value should be used.
- Presumptive Numbers of Heavy Vehicle Axle Groups per Heavy Vehicle ( $N_{HVAG}$ )

|             | $N_{HVAG}$ |
|-------------|------------|
| Rural Roads | 2.8        |
| Urban Roads | 2.5        |

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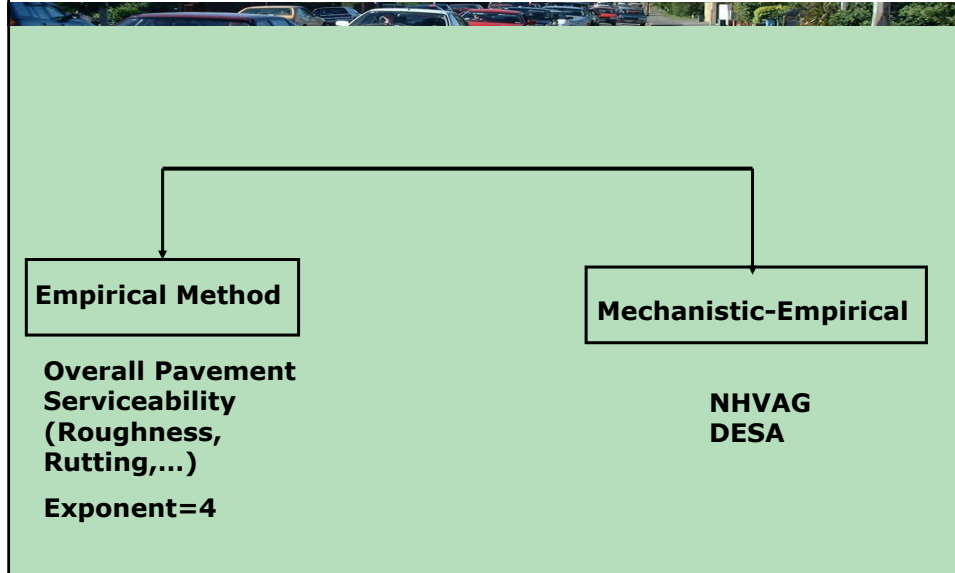
## Estimation of Traffic Load Distribution (TLD)

- TLD provides essential information to compute the damage caused by the HVAG. It contains:
  - Proportion of all axle Groups (SAST, SADT....)
  - For each Axle Group type, the proportion of axles applied at each load magnitude.

TLD can be used to calculate the average number of heavy axle groups per heavy vehicle ( $N_{HVAG}/HV$ ) and together with the SAR can be used to calculate the average number of ESA per axle group ( $ESA/HVAG$ ),  $SAR_a/ESA$ ,  $SAR_s/ESA$ ,  $SAR_c/ESA$ .

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# Damage to Flexible pavement



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# Pavement Damage in Terms of Standard Axle Repetitions

- The extent of damaged is assessed in terms of the number of load repetitions of the standard axle.
- The Standard Axle is a single axle with dual tyres (SADT) applying an axle load of 80 kN to the pavement.

$$SAR_{ijk} = \left( \frac{L_{ij}}{SL_i} \right)^m$$

**ONLY for Exponent 4**  
**Otherwise phased out**

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- $SAR_{ijk}$  = Number of Standard Axle Repetitions (or Passages of the Standard Axle) which causes the same damage of type k as a single passage of axle group type i with load  $L_{ij}$
- $SL_i$  = Standard Load for axle group type I
- $L_{ij}$  = Load on axle group
- k = Damage type
- m = an exponent which is specific to type k damage

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## Axle group Loads which cause the same damage as the standard axle

| Axle Group Type                              | Load (kN) |
|----------------------------------------------|-----------|
| Single axle with single tyres (SAST)         | 53        |
| Single axle with dual tyres (SADT)           | 80        |
| Tandem axle with single tyres (TAST)         | 90        |
| Tandem axle with dual tyres (TADT)           | 135       |
| Triaxle (Tridem) axle with dual tyres (TRDT) | 181       |
| Quad axle with dual tyres (QADT)             | 221       |

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## Changes in New Guidelines

- The concept of **SARa, SARs, SARc** has been phased out in Austroads 2017 subsequent editions, and the load equivalency concept can only expressed in terms of overall damage ESA
- Load Multipliers (SARa/ESA, SARc/ESA, SARs/ESA) are no longer used and in Circly use 1 for all of them.

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## Example

- Design Period = 20 years
- AADT = 5000
- Directional split factor = 0.5, and lane LDF= 1.0
- %HV= 8.0%
- $r = 4.0\%$
- $N_{HVAG}/HV = 2.4$
- Traffic Axle Load Distribution, see Table
- Determine the Design traffic.

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## Example

### Traffic Load Distribution

| Axle Group<br>Load (kN) | SAST<br>% | TADT<br>% | TRDT<br>% |   |
|-------------------------|-----------|-----------|-----------|---|
| 20                      | 10        | 5         | 10        |   |
| 70                      | 40        | 30        | 15        |   |
| 120                     | 50        | 15        | 20        |   |
| 340                     |           | 50        | 25        |   |
| 350                     |           |           | 30        |   |
| Proportion              | 0.4       | 0.4       | 0.2       | 1 |

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- $CGF = 29.78$
- $N_{HVAG} = 5,217,456$
- Use the traffic table to split NHVAG to different axle groups
- $SAST = 0.4 * 5217456 = 2,086,982$
- $TADT = 0.4 * 5217456 = 2,086,982$
- $TRDT = 0.2 * 5217456 = 1,043,491$

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## Design Traffic for Rigid Pavements

| Axle Group | SAST      | TADT    | TRDT    | Grand Total |
|------------|-----------|---------|---------|-------------|
| 20         | 208698    | 104349  | 104349  |             |
| 70         | 834793    | 626095  | 156524  |             |
| 120        | 1,043,491 | 313047  | 208698  |             |
| 340        | 0         | 1043491 | 260873  |             |
| 350        | 0         | 0       | 313047  |             |
| Total      | 2086982   | 2086982 | 1043491 | 5217455     |